

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Eighth Semester B.E. (Mechanical) (Credit System) Degree Examinations**

May - June 2023

19MEE26 / 17ME823 - FLUID POWER SYSTEMS

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

Unit – I

	Marks	BT*	CO*	PO*
1. a) With a neat sketch explain the basic components involved in a hydraulic system.	10	L*2	1	1
b) Analyze the cylinder loading system when the load rod pin lies between the cylinder rod pin and fixed hinge pin of the lever. Derive the expression for the given conditions.	5	L4	2	2
c) Explain tandem cylinder with sketch.	5	L4	2	2
2. a) A hydraulic jack used in an automobile has to lift a load of 5 kN by using an output cylinder having a diameter of 120 mm and input cylinder diameter of 40 mm. Calculate the input force to lift 5 kN load. Also determine how much an input cylinder has to move to raise the output cylinder by 100 mm. Sketch the above system.	5	L2	1	2
b) Explain briefly the different types of fluids used in hydraulic systems.	5	L2	1	1
c) A hydraulic motor used in packaging industry has a displacement of 300 cm ³ /rev and a speed of 2000 rpm with a pressure drop of 200 bar. The volumetric efficiency and mechanical efficiency of the motor are 90% and 95% respectively. Determine i) Theoretical and actual discharge ii) Theoretical and actual torque iii) Theoretical and actual power.	10	L3	2	2
3. a) Explain briefly the parameters to be considered during the selection of pump for hydraulic system applications.	4	L2	1	1
b) A vane pump consists of ten vanes each with 50 mm width. If the maximum cam ring diameter is 125 mm and rotor diameter is 100 mm, calculate the theoretical discharge of the pump. Consider the pump speed of 1200 rpm, eccentricity 9 mm, with a pressure rating of 69 bar and mechanical efficiency is 90%. Also calculate the theoretical and actual power required to drive the pump.	6	L3	1	2
c) Draw neat sketch of Telescopic type cylinder and explain its construction and working.	10	L2	2	1

Unit – II

4. a) Write the symbols to represent the following: i) 4-W, 2-P solenoid actuated spring centered DCV ii) Sequence valve iii) Pressure and temperature compensated flow control valve iv) Pressure relief valve.	4	L1	3	1
b) Sketch any four types of neutral positions used in four way three position direction control valves.	6	L2	3	1
c) With the help of a hydraulic circuit explain how to achieve sequence of operation (clamping and bending) by two cylinders in bending application?	10	L3	4	3

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| 5. a) | With the help of a neat sketch, explain the construction and working of pressure reducing valve. | 10 | L2 | 3 | 1 |
| b) | With the help of a hydraulic circuit explain, how a vertically mounted cylinder carrying heavy weight will be held in position against a gravitational force (Counter balancing) to safeguard the operator during loading and unloading of components in stamping operation? | 10 | L3 | 4 | 3 |
| 6. a) | With the help of neat sketch explain construction and working of pressure compensated flow control valve. | 10 | L2 | 3 | 1 |
| b) | Explain with a hydraulic circuit how an accumulator can be used as an emergency power source in hydraulic system applications. | 5 | L2 | 4 | 1 |
| c) | In a regenerative circuit "a small piston rod area provides a larger extension speed." Justify this statement with necessary equation and hydraulic circuit. | 5 | L3 | 4 | 1 |

Unit – III

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| 7. a) | With the help of a neat sketch explain the constructional features and working principles of a lubricator used in pneumatic system application. | 10 | L2 | 5 | 1 |
| b) | With the help of a neat sketch explain the constructional features and working principles of quick exhaust valve. | 10 | L2 | 5 | 1 |
| 8. a) | With the help of a block diagram, explain the basic structure of a pneumatic system. | 6 | L2 | 5 | 1 |
| b) | With the help of a pneumatic circuit explain the application of time delay valve. | 8 | L2 | 5 | 1 |
| c) | With the help of symbols, explain the applications of shuttle valve and two pressure valve used in pneumatic system applications. | 6 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
